

Analytical Problem Solving.

1) Gray Level Transformation

a) Gray Level Transformation is used to improve the visual quality of an image by modifying Pixel Intensity values. It enhance bright used contrast & makes important image details.

b) The Negative Transformation

given values

$$S = 255 - r$$

Pixel values

$$r = 50, 100, 150, 200$$

Now, $r = 50$

$$S = 255 - 50 = 205;$$

$$r = 100$$

$$S = 255 - 100 = 155;$$

$$r = 150$$

$$S = 255 - 150 = 105;$$

$$r = 200$$

$$S = 255 - 200 = 55;$$

Original Pixel	Negative Pixel
50	205
100	155
150	105
200	55

Interpret

Negative transformation convert dark pixels into bright pixels and bright pixels into dark pixels.

2) Histogram Equalization

a) Histogram equalization improves image contrast by redistributing the intensity value and the intensity range. It enhances the visibility of image details.

b) The cumulative distribution function [CDF]

Intensity	frequency
0	4
1	6
2	10
3	6

$$PDF = \frac{\text{count of the Pixel values}}{\text{Total No. of Pixel values}}$$

$$\text{Total No. of Pixel values} = 4 + 6 + 10 + 6 = 26$$

Now find out the PDF.

$$PDF(0) \Rightarrow \frac{4}{26} = 0.1538 \quad PDF(2) \Rightarrow \frac{10}{26} = 0.3846$$

$$PDF(1) \Rightarrow \frac{6}{26} = 0.2307 \quad PDF(3) \Rightarrow \frac{6}{26} = 0.2307$$

Here PDF's

$$P(0) = 0.153, \quad P(1) = 0.230, \quad P(2) = 0.384, \quad P(3) = 0.230$$

c) The equalized intensity

$$\text{Formula} = S = CDF \times 255$$

Now

$$P(0) \Rightarrow S = 0.153 \times 255 = 39.015$$

$$P(1) \Rightarrow S = 0.230 \times 255 = 58.65$$

$$P(2) \Rightarrow S = 0.384 \times 255 = 97.92$$

$$P(3) \Rightarrow S = 0.230 \times 255 = 58.65$$

∴

$$P(0) = 39.015$$

$$P(1) = 58.65$$

$$P(2) = 97.92$$

$$P(3) = 58.65$$

CDF?

Gaussian Smoothing

Gaussian smoothing filtering reduce image noise by replacing with a weighted average of its neighbours. Give a long higher weight than surrounding pixels. so it preserves image details better and produces.

The Gaussian kernel.

$$\text{given } \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Gaussian kernel $\times$ image region

$$\Rightarrow \frac{(1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10)}{16}$$

$$\Rightarrow \frac{10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10}{16}$$

$$\Rightarrow \frac{360}{16} = 22.5$$

The final Answer = 22.5

Edge Detection using Sobel Operator.

Edge detection identifies object boundaries by detecting sudden changes in image intensity. The Sobel Operator calculates image gradients to detect horizontal and vertical edges.

2, by Horizontal Sobel mask
 a) given

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

the image

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

b)

$$\Rightarrow \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$\Rightarrow (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + 0 + 0 + 0 + (1 \times 30) + (2 \times 30) + (1 \times 30)$$

$$\Rightarrow -10 + (-20) + (-10) + 30 + 60 + 30$$

$$\Rightarrow -40 + 120$$

$$\Rightarrow 80$$

The final Answer = 80.

The gradient at the center and interpret the result is 80.

5

Contrast stretching.

Contrast stretching improves image quality by increasing the range of intensity values. It makes dark regions darker and bright regions brighter, improving the visibility of image details.

given

map the range $[50, 150]$ to $[0, 255]$

$$x = 50, 100, 150$$

Intensity Values in the range 50-150

formula 3

$$S = \frac{(x - x_{\min})}{x_{\max} - x_{\min}} \times 255$$

$$x_{\min} = 50$$

$$x_{\max} = 150$$

for 50

$$S = \frac{50 - 50}{150 - 50} \times 255 = \frac{0}{100} = 0$$

$$x = 100$$

$$S = \frac{100 - 50}{150 - 50} \times 255 = \frac{50}{100} \times 255 = 127.5$$

$$x = 150$$

$$S = \frac{150 - 50}{150 - 50} \times 255 = \frac{100}{100} \times 255 = 255$$

Original intensity values	Transformed intensity values
50	0
100	127.5
150	255